

Matheus Gomes Cordeiro

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SKILLS

Programming Languages: Python, R, SQL, C/C++, MATLAB/Octave, Wolfram Mathematica

Technologies & Tools: Git, GitHub, GitLab, Databricks, RabbitMQ, Elasticsearch, TensorFlow/Keras, scikit-learn, PyTorch, NLTK, spaCy, statsmodels, NeuralForecast by Nixtla, Prophet, PySpark, python-bcb, BacenAPI, sktime, NumPy, Numba, Cython, mpmath, pytest, pip, pandas, pymove, Matplotlib, Plotly, Seaborn, Voilà, Panel, Streamlit, pdoc, Transformers/Hugging Face, PostgreSQL, MongoDB, Jupyter Notebook, Google Colab, LaTeX, Markdown

PROFESSIONAL EXPERIENCE

University of Fortaleza (Unifor), Fortaleza, Brazil

Jul 2026 – Present

Assistant Professor – Part-Time

- Teach on-site undergraduate courses in Mathematics for Computing, Problem Modeling with Mathematics, Logical and Algorithmic Reasoning, and Collaborative Cyber-Physical Systems.
- Develop students' foundations in mathematical reasoning, computational thinking, algorithmic problem-solving, and software–hardware integration. Topics include logic and proof methods, set theory, functions, sequences, recurrence relations, calculus and numerical methods, programming fundamentals, data structures and algorithms, embedded and distributed systems, sensors, actuators, and the Internet of Things (IoT).
- Supervise hands-on activities involving ESP32-based systems, serial communication, Wi-Fi, MQTT, ESP-NOW, sensor integration, and IoT prototyping.
- Plan lectures, learning activities, and assessments and supervise practical projects, connecting mathematical, algorithmic, software, and hardware concepts to computational problem-solving and real-world applications.

ACT Digital

Aug 2025 – 2026

Data Scientist II

- Bank Budget Forecasting:
 - Developed AI-driven forecasts of credit-loss provisions based on the monthly Expected Credit Loss percentage (ECL%) to support compliance with Central Bank of Brazil (BACEN) Resolution No. 4,966.
- Credit Model Validation Agent MVP:
 - Worked as a consultant on the development of a Minimum Viable Product (MVP) for an AI agent designed to test credit models' adherence to defined validation criteria, in collaboration with Domo, a third-party company.
- Standardized Credit Model Validation Framework:
 - Led the technical design, at Banco BMG, of a proposal for a standardized credit model validation pipeline structured as a reusable framework.
 - Designed a scoring process to quantitatively assess validation criteria and outcomes on a consistent scale, improving consistency, comparability, and traceability across evaluated models.
 - Mapped validation scores to qualitative governance outcomes — *Accepted*, *Accepted with Reservations*, or *Rejected* — using predefined score ranges to support model approval decisions.
- Tools and Methodologies:
 - Applied time-series and machine-learning models, including ARIMA/ARIMAX, SARIMA/SARIMAX, AutoARIMA, N-BEATS, N-HITS, Autoformer, FEDformer, Prophet, KNeighborsRegressor, and XGBRegressor.
- Technologies:
 - Developed technical solutions in R (RStudio) and Python using NumPy, pandas, scikit-learn, Matplotlib, statsmodels, sktime, NeuralForecast by Nixtla, Prophet, PySpark, python-bcb, and BacenAPI.
 - Used SQL and Databricks for data processing, analysis, and management.
- Client:
 - Banco BMG.

Data Scientist II

- i) Risk Model Validation through VOYAGER IT Innovative Solutions as an outsourced consultant:
 - Validated four models covering Credit Risk, Microcredit Risk, Operational Risk, Market Risk, and Social, Environmental, and Climate Risks.
 - Ensured model compliance and governance by reviewing documentation against internal policies, external regulations, and governance frameworks.
 - Performed advanced technical analyses, including:
 - Exploratory Data Analysis (EDA) to identify relevant patterns and insights.
 - Data preprocessing to improve the quality and reliability of the information used by the models.
 - Reproduction of existing models to verify the accuracy and consistency of their results.
 - Development of challenger models to test and improve predictive performance.
 - Model performance assessment for benchmarking and continuous improvement.
- ii) Tools and Methodologies:
 - Applied supervised machine-learning algorithms, including MLP, XGBoost, LightGBM, and XGBSE, as well as unsupervised techniques such as K-means, PAM, and CLARA.
 - Used statistical modeling methods, including Value at Risk (VaR), rule-based systems, and simplified mathematical models such as RAROC.
- iii) Developed technical solutions in R (RStudio), Python using NumPy, pandas, scikit-learn, Matplotlib, and TensorFlow, and SQL.

Quantum Optics and Information Group (GIQ), Fortaleza, Brazil

Feb 2023 – Feb 2025

Quantum Computing Simulation Researcher

- Conducted research on the accurate and efficient simulation of the Quantum Harmonic Oscillator, a widely used model in photonic quantum computing.
The research resulted in the development of a Python package named **Fast Wave**. The work was co-advised by Dr. Scott Glancy from the National Institute of Standards and Technology (NIST):
Fast Wave GitHub Repository ,
Fast Wave Documentation .
- Python 3, Numba, Cython, mpmath, pytest, pdoc, pip, Google Colab, Git, and GitHub.

SeaFortress, Fortaleza, Brazil

Jan 2018 – Present

Chief Artificial Intelligence Officer – CAIO

- Co-founder and Chief Artificial Intelligence Officer of **SeaFortress**, a startup operating under a hybrid business model that combines an independent game studio with a software development company.
Led the application of Artificial Intelligence across independent game development and custom virtual experiences, focusing on accessible and cost-efficient solutions.
Contributed directly to **Sentis**, Unity's deep-learning inference library, and had a SeaFortress game showcased at **GDC 2024**:
SeaFortress Website .
- Python 3, Transformers/Hugging Face, PyTorch, C#, Sentis, and Unity.

Insight Data Science Lab, Fortaleza, Brazil

Jan 2022 – Jan 2023

Artificial Intelligence Researcher

- Conducted research combining Natural Language Processing (NLP) and vehicle trajectory analysis.
Trained BERT-based models from the Hugging Face Transformers library on vehicle trajectory datasets to assess whether the next trajectory element could be predicted similarly to next-word prediction in a text sequence.
The research resulted in the development of a dashboard application named **Teach**:
Teach GitHub Repository .
- Python 3, Transformers/Hugging Face, PyTorch, pandas, pymove, Voilà, Jupyter Notebook, and GitHub.

LEAD Dell, Fortaleza, Brazil

Jul 2021 – Jan 2022

Artificial Intelligence Developer

- Worked as part of a Research and Development team responsible for building a machine-learning application to detect anomalies in Dell systems through system logs.
Integrated the Isolation Forest model from scikit-learn into a pipeline that received logs from RabbitMQ, processed the data, and stored the results in Elasticsearch.
- Python 3, scikit-learn, RabbitMQ, and Elasticsearch.

EDUCATION

Federal University of Ceará (UFC), Fortaleza, Brazil

Feb 2023 – Feb 2025

Master of Science in Teleinformatics Engineering

Main research areas: Scientific Computing, Computational Physics, Quantum Computing Simulation, Quantum Devices and Systems, and Quantum Optics.

Federal University of Ceará (UFC), Fortaleza, Brazil

Jan 2017 – Jan 2023

Bachelor of Science in Computer Engineering

Relevant coursework: Linear Algebra; Differential, Integral, Vector, and Numerical Calculus; Ordinary Differential Equations; Probability and Statistics; Data Mining; Programming Logic; Object-Oriented Programming; Data Structures; Image Processing; Graph Algorithms; Graph Theory; and Neural Networks.

PUBLICATIONS

- [1] Cordeiro, M., Bezerra, I. P., & Vasconcelos, H. H. M. (2024). Efficient computation of the wave function $\psi_n(x)$ using Hermite coefficient matrix in Python. In *Proceedings of the 7th Workshop-School on Quantum Computing and Information (7th WECIQ)* (pp. 56–60). Rio de Janeiro: CEFET/RJ.
- [2] Cordeiro, M., Markert, C., Araújo, S. S., Campos, N. G. S., Gondim, R. S., da Silva, T. L. C., & da Rocha, A. R. (2022). Towards smart farming: Fog-enabled intelligent irrigation system using deep neural networks. *Future Generation Computer Systems*, 129, 115–124. <https://doi.org/10.1016/j.future.2021.11.013>
- [3] Cordeiro, M. G., Serafim, P. B. S., Nogueira, Y. L. B., Vidal, C. A., & Cavalcante Neto, J. B. (2019). A minimal training strategy to play Flappy Bird indefinitely with NEAT. In *2019 18th Brazilian Symposium on Computer Games and Digital Entertainment (SBGames)* (pp. 21–28). <https://doi.org/10.1109/SBGames.2019.00014>
- [4] Cruz, L. A., Coelho da Silva, T. L., Magalhães, R. P., Melo, W. C. D., Cordeiro, M., de Macedo, J. A. F., & Zeitouni, K. (2022). Modeling Trajectories Obtained from External Sensors for Location Prediction via NLP Approaches. *Sensors*, 22(19), 7475. <https://doi.org/10.3390/s22197475>
- [5] Freitas, N. A. de, da Silva, T. C., de Macêdo, J. F., Melo Junior, L., & Cordeiro, M. (2021). Using deep learning for trajectory classification. In *Proceedings of the 13th International Conference on Agents and Artificial Intelligence – Volume 2: ICAART* (pp. 664–671). SciTePress. <https://doi.org/10.5220/0010227906640671>

CERTIFICATIONS

Learning Tracks:

- Data Analysis in Databricks — DataCamp

Professional Courses:

- Data Management in Databricks — DataCamp
- Databricks Concepts — DataCamp
- Introduction to Databricks — DataCamp
- Data Visualization in Databricks — DataCamp
- Credit Risk Modeling in Python — DataCamp

Professional Accreditations:

- Academy Accreditation – Databricks Fundamentals
- NVIDIA Fundamentals of Deep Learning

Academic and Scientific Credentials:

- Certificate of Publication — Sensors
- Certificate of Publication — Future Generation Computer Systems